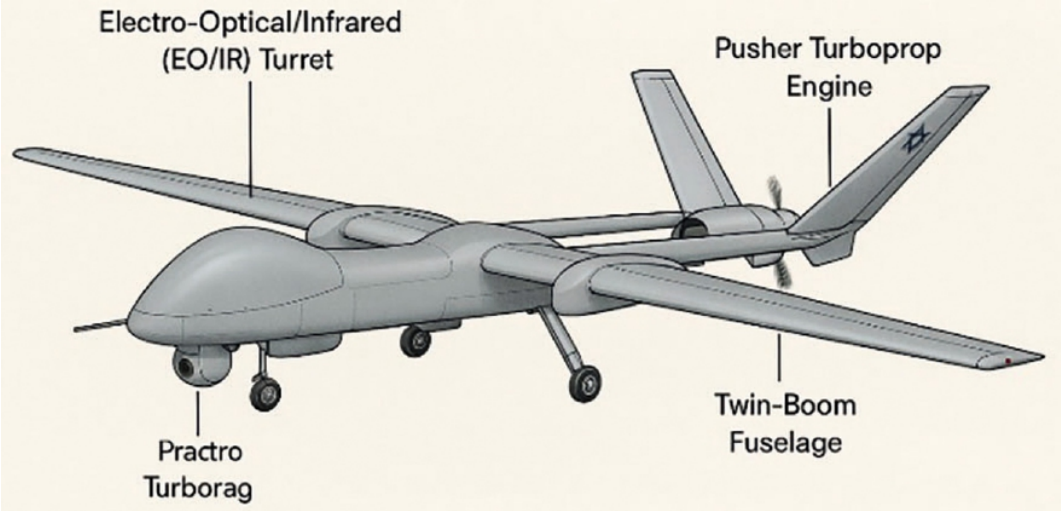


Figure 4.5 Heron TP (Eitan) (2004).⏎

4.5.2.5 Dassault Neuron (2006)

The Dassault Neuron, unveiled in 2006, represents a landmark achievement in the development of stealth and autonomous combat UAVs (UCAVs), marking the transition toward fourth-generation unmanned systems. Developed by Dassault Aviation (France) under a Pan-European collaboration including Italy (Alenia Aermacchi), Sweden (Saab), Spain (EADS-CASA), Greece (HAI), and Switzerland (RUAG), the Neuron project was the first European stealth UCAV demonstrator. Its purpose was to explore autonomous flight, low-observable design, and precision strike capability while validating advanced aerodynamics, sensor fusion, and control systems for future unmanned combat air systems (UCAS).

Design and technical overview

The Neuron is a flying-wing UCAV designed with an emphasis on stealth, autonomy, and precision strike. Its blended body-wing configuration, crafted from carbon-fiber composites, minimizes radar cross-section (RCS) and enhances aerodynamic efficiency. With a wingspan of 12.5 meters (41 feet) and a length of 9.2 meters (30 feet), the aircraft achieves a balanced design between agility, endurance, and payload capability.